Solution: Target Sum

Problem Statement

Given a set of positive numbers (non zero) and a target sum 'S'. Each number should be assigned either a '+' or '-' sign. We need to find out total ways to assign symbols to make the sum of numbers equal to target 'S'.

Example 1:

```
Input: {1, 1, 2, 3}, S=1
Output: 3
Explanation: The given set has '3' ways to make a sum of '1': {+1-1-2+3}
& {-1+1-2+3} & {+1+1+2-3}
```

Example 2:

```
Input: {1, 2, 7, 1}, S=9
Output: 2
Explanation: The given set has '2' ways to make a sum of '9': {+1+2+7-1}
& {-1+2+7+1}
```

Solution

This problem follows the **0/1 Knapsack pattern** and can be converted into `"Count of Subset Sum"`. Let's dig into this.

We are asked to find two subsets of the given numbers whose difference is equal to the given target 'S'. Take the first example above. As we saw, one solution is {+1-1-2+3}. So, the two subsets we are asked to find are {1, 3} & {1, 2} because,

```
(1 + 3) - (1 + 2 ) = 1
```

Now, let's say 'Sum(s1)' denotes the total sum of set 's1', and 'Sum(s2)' denotes the total sum of set 's2'. So the required equation is:

```
Sum(s1) - Sum(s2) = S
```

This equation can be reduced to the subset sum problem. Let's assume that 'Sum(num)' denotes the total sum of all the numbers, therefore:

```
Sum(s1) + Sum(s2) = Sum(num)
```

Let's add the above two equations:

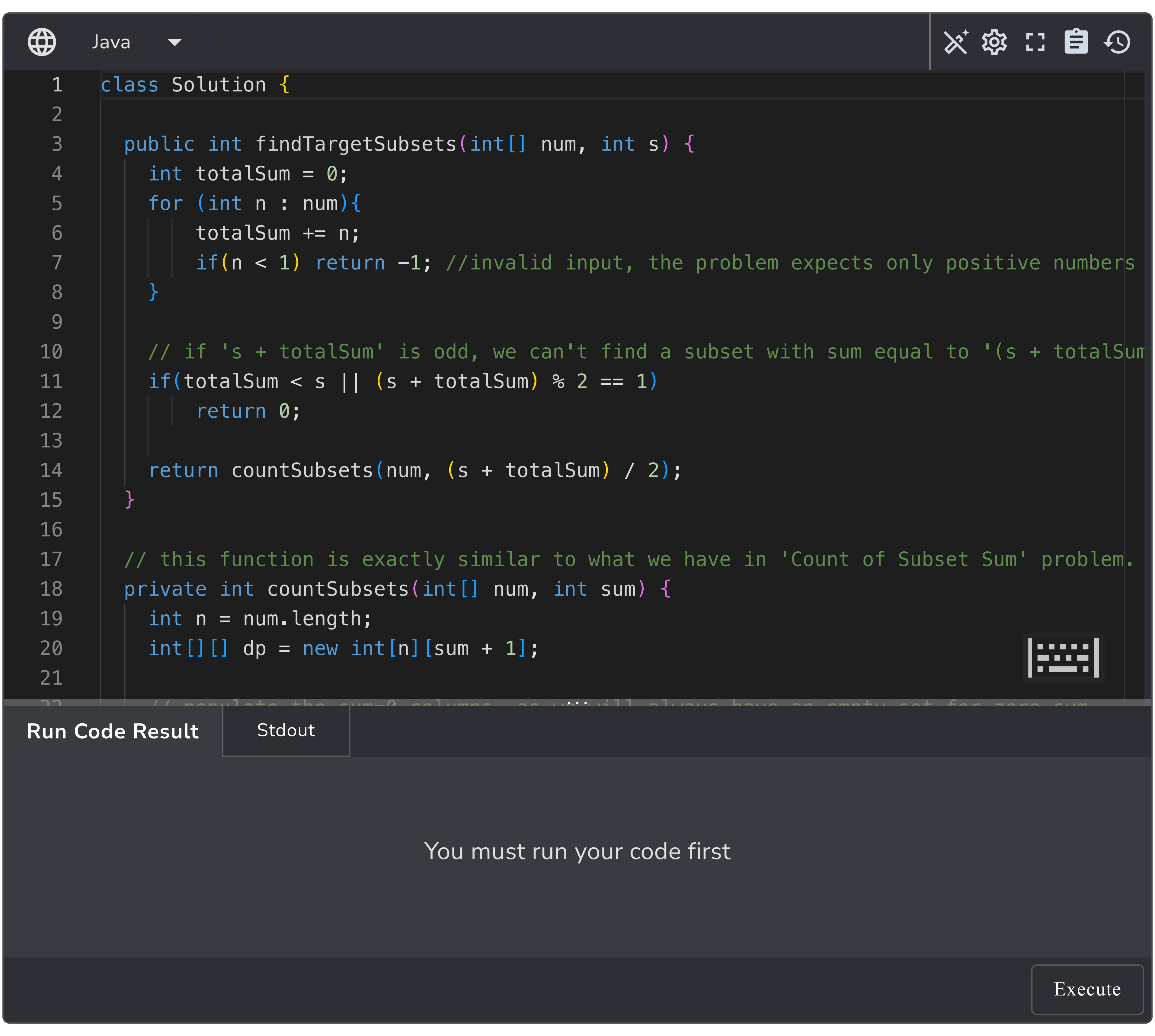
```
=> Sum(s1) - Sum(s2) + Sum(s1) + Sum(s2) = S + Sum(num)
=> 2 * Sum(s1) =  S + Sum(num)
=> Sum(s1) = (S + Sum(num)) / 2
```

This essentially converts our problem to: "Find count of subsets of the given numbers whose sum is equal to",

```
=> (S + Sum(num)) / 2
```

Code

Let's take the dynamic programming code of `"Count of Subset Sum"` and extend it to solve this problem:



```java
class Solution {

  public int findTargetSubsets(int[] num, int s) {
    int totalSum = 0;
    for (int n : num){
      totalSum += n;
      if(n < 1) return -1; //invalid input, the problem expects only positive numbers
    }

    // if 's + totalSum' is odd, we can't find a subset with sum equal to '(s + totalSum
    if(totalSum < s || (s + totalSum) % 2 == 1)
      return 0;

    return countSubsets(num, (s + totalSum) / 2);
  }

  // this function is exactly similar to what we have in 'Count of Subset Sum' problem.
  private int countSubsets(int[] num, int sum) {
    int n = num.length;
    int[][] dp = new int[n][sum + 1];

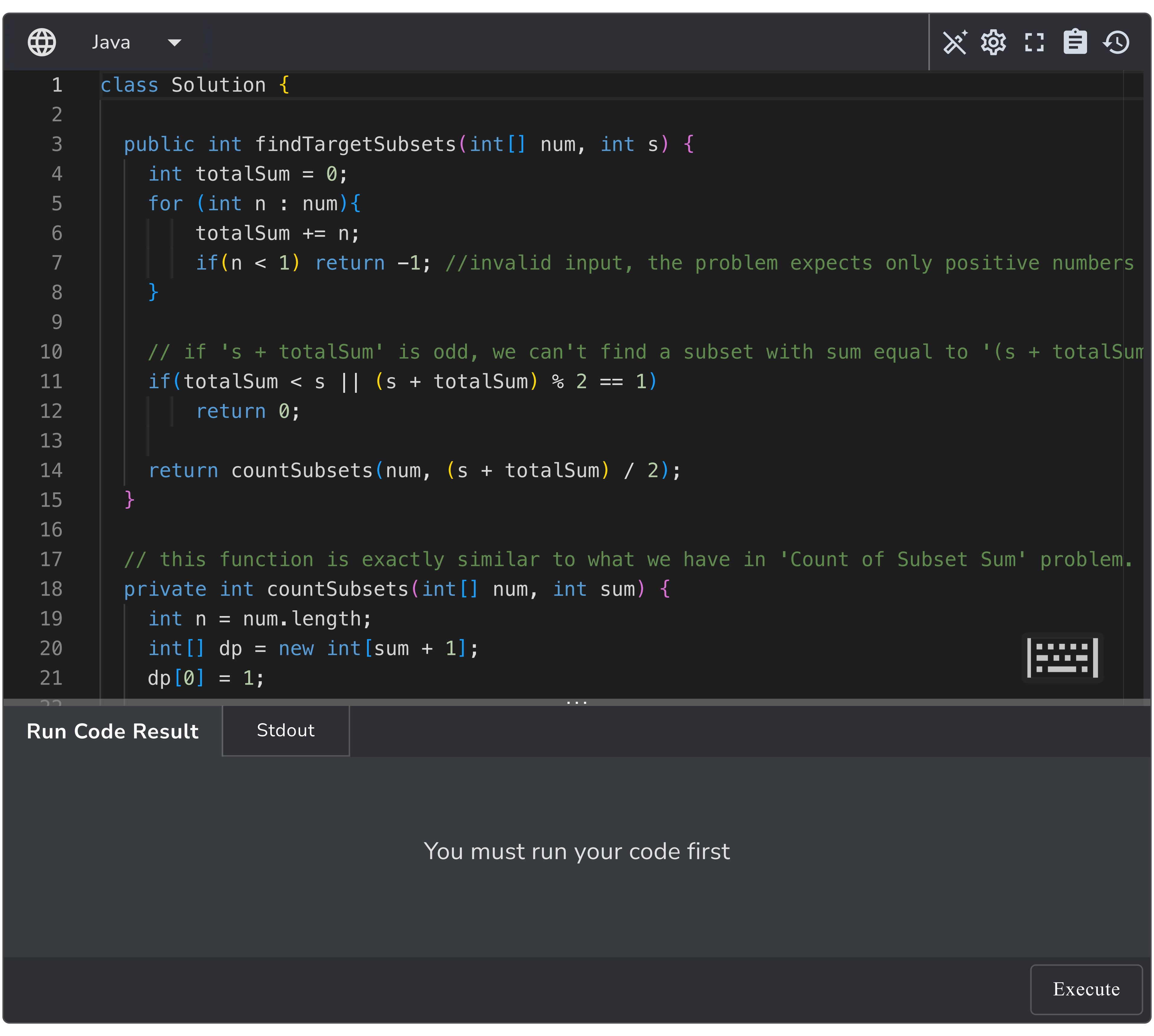
```

The above solution has time and space complexity of $O(N * S)$, where 'N' represents total numbers and 'S' is the desired sum.

We can further improve the solution to use only $O(S)$ space.

Space Optimized Solution

Here is the code for the space-optimized solution, using only a single array:



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    if(totalSum < s || (s + totalSum) % 2 == 1)
      return 0;

    return countSubsets(num, (s + totalSum) / 2);
  }

  // this function is exactly similar to what we have in 'Count of Subset Sum' problem.
  private int countSubsets(int[] num, int sum) {
    int n = num.length;
    int[] dp = new int[sum + 1];
    dp[0] = 1;
```